TQUK Functional Skills Qualification in Maths at Level 2

Examination Past Paper 1

Please complete the details below using black or blue ink. Use BLOCK CAPITALS.

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

Instructions:

- Read each question **carefully**
- Answer **all** questions
- Write your answers **clearly** in the spaces provided
- **Check** your answers.

Information:

- This examination has **two** sections. These are clearly labelled
- You are **not allowed** to use a calculator for Section A
- You **are allowed** to use a basic calculator for Section B
- The **maximum** mark for this examination is 60
- The marks available for each question are shown in **bold** beneath each question.

Items:

- You **will need** a pen with black or blue ink, a pencil, a ruler and an eraser (for diagrams, graphs and charts only)
- You **will need** a basic calculator for Section B only
- You **will need** a protractor and a compass
- You **will not** need any other stationery or equipment.

Time allowed:

30 minutes for Section A (Non-calculator)
90 minutes for Section B (Calculator)

Do not open this examination paper until you are told to do so.

For examiner use only

	Marks available	Marks awarded	Second marks
Section A	15		
Section B	45		
Total marks	60		

Past paper

© Training Qualifications UK 2023 All Rights Reserved

This page is intentionally blank.

Past paper

© Training Qualifications UK 2023 All Rights Reserved

Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

1. Work out 205 983 + 826 148

Answer	

[1]

2. Work out 2.5 × 3.462

Answer	

[1]

Past paper

Questions continue on the following page

© Training Qualifications UK 2023 All Rights Reserved

3. Charlie buys a length of wood equal to 3.6 metres (m).

Charlie cuts off a length of 1.125 metres (m).

Work out how much wood Charlie has left.

Give your answer in metres (m).

	Answer	**m**

[2]

4. Work out $7^2 + 6 \times 5$

	Answer	

[2]

Past paper

© Training Qualifications UK 2023 All Rights Reserved

5. Harper wants to go on holiday.

Harper finds two offers for the same holiday on different websites:

Website A	**Website B**
38% off	$\frac{3}{8}$ off

Which offer gives the bigger discount?

Show how you decide.

Answer	

[2]

Past paper

Questions continue on the following page

© Training Qualifications UK 2023 All Rights Reserved

6. 60 people attend a dancing school.

The classes each person attends are shown below:

	Morning	**Afternoon**
Ballet	28	15
Tap	12	5

One person is chosen at random.

What is the probability that the person chosen attends the morning ballet class?

Give your answer as a fraction in its simplest form.

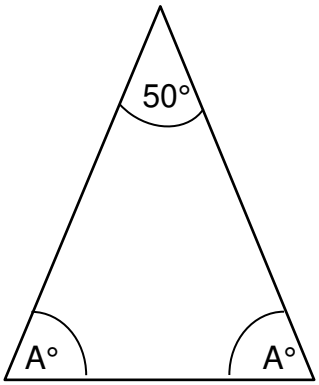
Answer	

[2]

Past paper

© Training Qualifications UK 2023 All Rights Reserved

7. Work out the value of A shown in this triangle:



Not drawn to scale

Answer		°

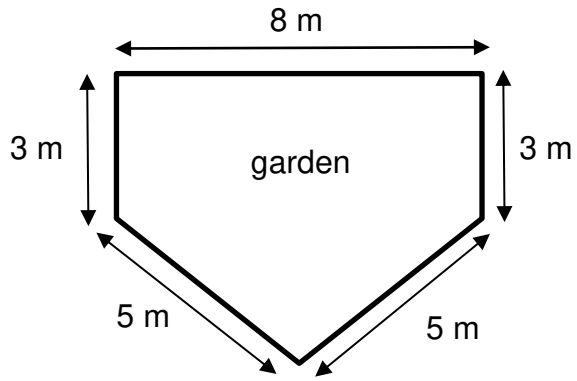
[2]

Past paper

Questions continue on the following page

© Training Qualifications UK 2023 All Rights Reserved

8. Drew wants to put a fence around a garden.

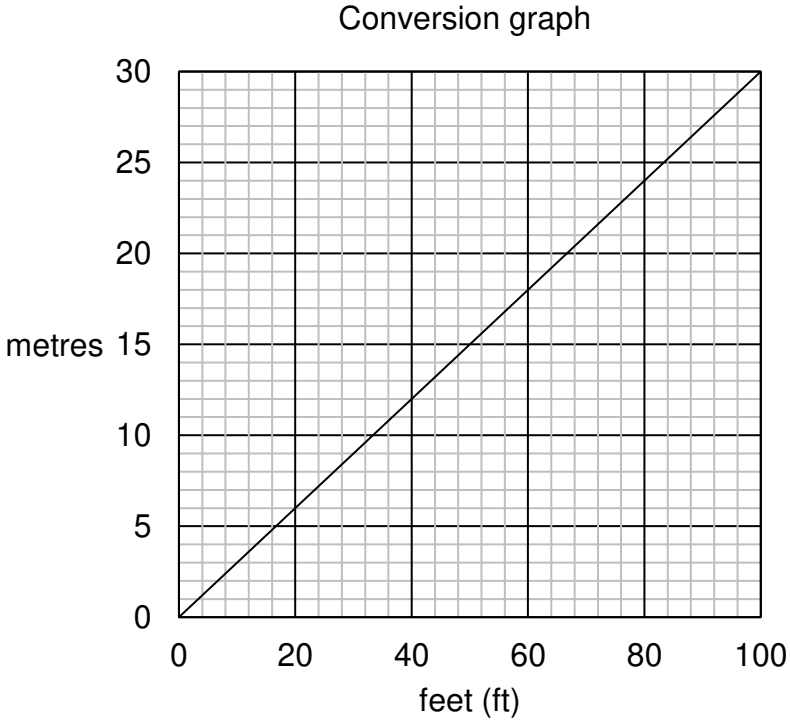


Not drawn to scale

Drew knows the fence panels are sold in feet (ft).

How many feet (ft) of fence panelling does Drew need in total?

Use this conversion graph:



Answer	**ft**

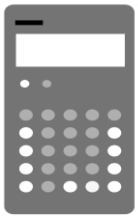
[3]

End of Section A.

Section B begins on Page 8.

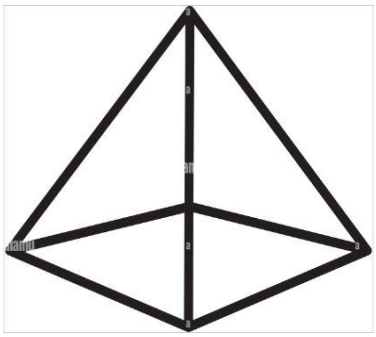
© Training Qualifications UK 2023 All Rights Reserved

Section B: Calculator

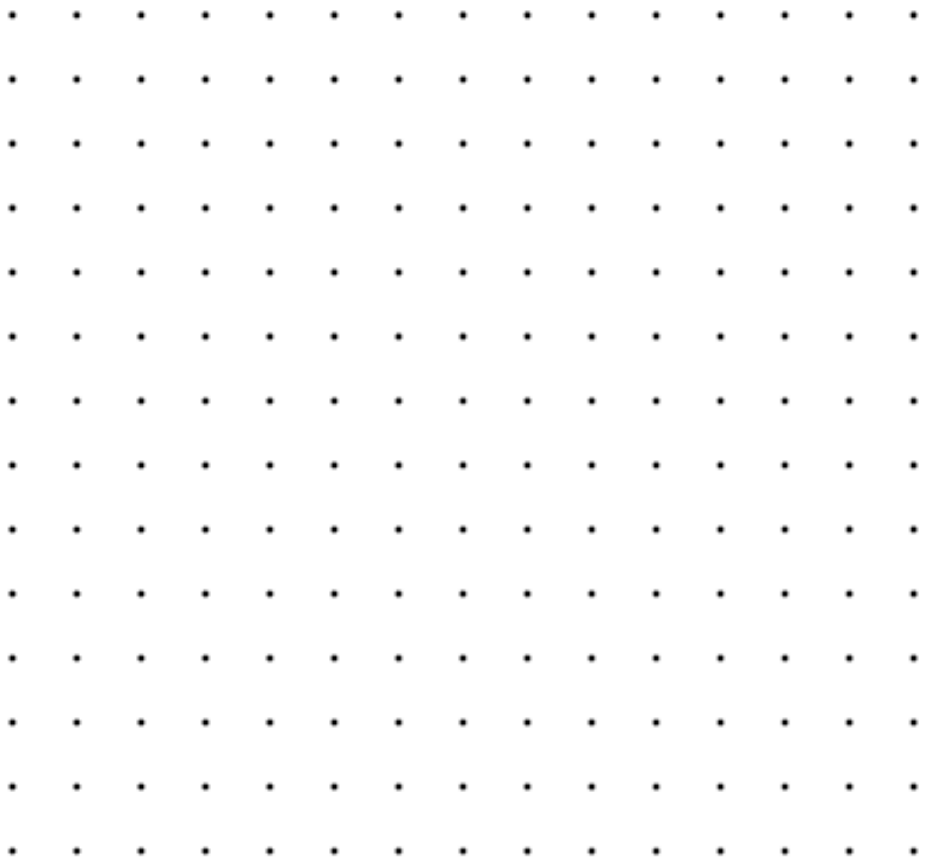


There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

1. The diagram shows a square-based pyramid:



Draw the plan view of the square-based pyramid on this grid:



[1]

Paper

© Training Qualifications UK 2023 All Rights Reserved

2. Write three billion in numbers.

Answer	

[1]

3. Lee wants to buy a new computer for £1390

If Lee uses a payment plan, the price of the computer increases by 12.5%

How much money will Lee pay in total for the computer using the payment plan?

	Answer	**£**

[2]

Past paper

Questions continue on the following page

© Training Qualifications UK 2023 All Rights Reserved

4. Put these fractions in order starting with the lowest:

$\frac{6}{7}$ $\frac{6}{5}$ $\frac{8}{7}$ $\frac{36}{35}$

Show any working:				
Answer				
	Lowest	→		**Highest**

[2]

5. 140 people were asked if they could speak a second language.

126 people said yes.

What percentage of the total number of people said yes?

	Answer	**%**

[2]

Past paper

© Training Qualifications UK 2023 All Rights Reserved

6. Harrison needs their new house to be painted within 8 days.

3 painters will take 5 days to paint the house.

Unfortunately, one painter is unavailable.

Will 2 painters be able to paint the house in time?

Show how you decide.

Answer	

[2]

7. Some friends are planning a hike.

The total distance of the hike is 105 kilometres (km).

The hiking route is divided into Sections A, B and C in the ratio 3 : 7 : 11

One friend thinks that Section B is 40 kilometres (km) longer than Section A.

Is this correct?

Show how you decide.

Answer	

[3]

Questions continue on the following page

Past paper

© Training Qualifications UK 2023 All Rights Reserved

8. Jude drives 136 kilometres (km) to visit a friend.

The drive takes 2.25 hours.

Jude thinks the average speed was less than 60.5 kilometres per hour (kph).

Is Jude correct?

Show how you decide.

Answer	

[3]

9. Sam needs 2.5 pounds (lbs) of flour to do some baking.

The shop sells flour in kilogram (kg) bags.

Sam buys a 2-kilogram bag of flour.

How much flour will Sam have left over?

Use 1 lb = 0.454 kg

Answer	

[3]

Past paper

© Training Qualifications UK 2023 All Rights Reserved

10. Kai gets paid a net amount of £1418.50 each month.

29% is deducted from Kai's gross pay before they are paid the net amount.

Kai wants to check the gross amount before the 29% was deducted.

Kai rounds the net amount to the nearest £10 and uses the rounded value to check the gross amount.

What answer should Kai get?

Show your working.

Answer	**£**

[3]

Past paper

Questions continue on the following page

© Training Qualifications UK 2023 All Rights Reserved

11. Maddison uses this water bottle whilst at the gym.

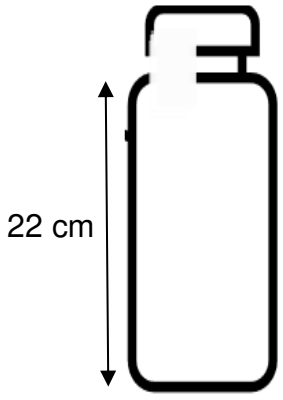
The bottle is in the shape of a cylinder with internal dimensions:

- Radius = 4 cm
- Height = 22 cm

Maddison's personal trainer recommends a new water bottle that holds 8% more water.

What size water bottle should Maddison buy?

Use $\pi = 3.14$



Not drawn to scale

Answer	**cm^3**

[4]

Past paper

© Training Qualifications UK 2023 All Rights Reserved

12. Two years ago, Nicky paid £8000 into a new savings account.

The savings account paid 4% compound interest per year.

Nicky sees this car for sale:

Car for sale
£12 975

All the money in the savings account will be used as a deposit towards the car.

The balance will be paid in 20 equal monthly instalments.

How much will each monthly instalment be?

Answer	**£**

[4]

Past paper

Questions continue on the following page

© Training Qualifications UK 2023 All Rights Reserved

13. Riley grows apples and oranges.

The weights of the apples collected on Monday are shown below:

173 grams	151 grams	182 grams	248 grams	214 grams

The weights of the oranges collected on Monday are summarised below:

Oranges weights	
Mean weight	188 grams
Range	112

Riley claims:

a) 'On average the weight of the apples is greater than the weight of the oranges.'

b) 'The weights of the apples are more consistent.'

Are each of Riley's claims correct?

Give reasons for your answers.

Show your working.

Show working here:
Reason - claim a)
Reason - claim b)

[4]

Past paper

© Training Qualifications UK 2023 All Rights Reserved

14. Taylor is going to Finland for six months and wants to know the average monthly temperature.

Taylor finds this information:

Month	November	December	January	February	March	April
Average temperature °F	32	29	28	30	35	46

Taylor uses this formula to convert the median temperature from degrees Fahrenheit (°F) to degrees Celsius (°C):

$$C = \frac{5(F - 32)}{9}$$

Where C = temperature in Celsius (°C)
F = temperature in Fahrenheit (°F)

Taylor thinks the median monthly temperature is colder than –1°C.

Is Taylor correct?

Show how you decide.

Answer	

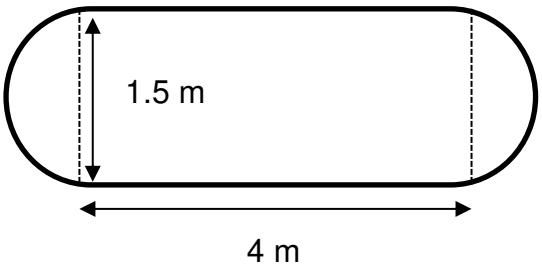
[5]

Questions continue on the following page

Past paper

© Training Qualifications UK 2023 All Rights Reserved

15. Alex has a table in the shape of a rectangle with a semicircle on each end.



Not drawn to scale

Alex wants to cover the surface of the table in fabric and finds these prices online:

Fabric Price per square metre						
£3.75	£2.99	£3.75	£4.00	£5.25	£4.99	£5.75

Alex uses the modal price to estimate how much it will cost to cover the surface in fabric.

How much does Alex estimate the fabric will cost?

Give your answer to the nearest whole pound (£).

Use π = 3.14

Answer	**£**

[6]

Past paper

This is the end of the assessment.

Past paper